

A disproof of Conjecture 00000001305

The Kobayashi pseudodistance of $\mathbb{C}^2 \setminus \mathbb{Z}^2$ vanishes identically

1 The conjecture and its reading

Conjecture 00000001305 reads: *Definition:* the Kobayashi metric d_K . *Conjecture:* on the domain in $\mathbb{C}^2$ whose complement is the lattice $\mathbb{Z}^2$ (an “Eisenstein complement”), d_K degenerates in the horizontal direction but not in the vertical direction (mixed hyperbolicity).

Let $D = \mathbb{C}^2 \setminus \mathbb{Z}^2$, where $\mathbb{Z}^2 = \{(m, n) : m, n \in \mathbb{Z}\} \subset \mathbb{C}^2$. We read d_K as the Kobayashi pseudodistance of D . “Degenerates in a direction” means: $d_K(p, q) = 0$ for all $p, q \in D$ differing only in that coordinate; “does not degenerate” means $d_K(p, q) > 0$ for all distinct $p, q \in D$ differing only in the other coordinate. Since the text does not say which coordinate is horizontal, we treat both assignments (Reading A: first coordinate horizontal; Reading B: the reverse). We prove more: $d_K \equiv 0$ on $D \times D$, so no reading that asks for positive d_K between some pair of distinct points of D can hold.

2 Definitions (as formalised in Lean)

Definition 1 (Poincare distance). For a, b in the open unit disc Δ , $\rho(a, b) = \operatorname{artanh} \left| \frac{a-b}{1-\bar{a}b} \right| = \frac{1}{2} \log \frac{1+x}{1-x}$ with $x = \left| \frac{a-b}{1-\bar{a}b} \right|$ (curvature -4 normalisation; another normalisation multiplies ρ by a positive constant and changes neither vanishing nor positivity).

Definition 2 (Kobayashi pseudodistance). A *chain* from p to r in D is a finite sequence $p = p_0, p_1, \dots, p_k = r$ together with holomorphic maps $f_j : \Delta \rightarrow D$ and points $a_j, b_j \in \Delta$ with $f_j(a_j) = p_{j-1}$, $f_j(b_j) = p_j$; its length is $\sum_j \rho(a_j, b_j)$ (the empty chain, $p = r$, has length 0). $d_K(p, r)$ is the infimum of the lengths of all chains, in $[0, \infty]$.

In Lean (`Conjecture1305/Basic.lean`): `C1305.D`, `C1305.poincare`, `C1305.IsDisc`, the inductive predicate `C1305.Chain`, and `C1305.dK` (an infimum in `ENNReal`). The holomorphy of a disc is `DifferentiableOn` $\mathbb{C}$ on the open unit ball, as a map $\mathbb{C} \rightarrow \mathbb{C} \times \mathbb{C}$.

3 Proof

Lemma 3 (one link). *Let $f : \mathbb{C} \rightarrow \mathbb{C}^2$ be entire with $f(\mathbb{C}) \subset D$, $f(0) = p$, $f(1) = q$, and let $R > 1$. Then $d_K(p, q) \leq \operatorname{artanh}(1/R)$.*

Proof. $g(z) = f(Rz)$ maps Δ holomorphically into D , $g(0) = p$, $g(1/R) = q$, and $\rho(0, 1/R) = \operatorname{artanh}(1/R)$. The one-link chain has that length. (Lean: `dK_1e`.) $\square$

Lemma 4 (entire curves). *For distinct $p, q \in D$ there is an entire $f : \mathbb{C} \rightarrow D$ with $f(0) = p$, $f(1) = q$.*

Proof. Write $u = q - p$. If $u_1 \neq 0$ put $f_s(t) = (p_1 + tu_1, p_2 + tu_2 + st(t - 1))$. For $t \in \{0, 1\}$, $f_s(t) \in \{p, q\} \subset D$. For $t \notin \{0, 1\}$, if $f_s(t) = (m, n) \in \mathbb{Z}^2$ then $t = (m - p_1)/u_1$ is determined by m and $s = (n - p_2 - tu_2)/(t(t - 1))$ is determined by (m, n) . So the bad s lie in a countable set, and since $\mathbb{C}$ is uncountable some s is good. If $u_1 = 0$ then $u_2 \neq 0$; exchange the two coordinates (D is invariant under the swap) and repeat. (Lean: `core, exists_curve`.) $\square$

Theorem 5. $d_K(p, q) = 0$ for all $p, q \in D$.

Proof. If $p = q$ the empty chain gives 0. Otherwise, for $\varepsilon > 0$ pick $R = 1 + 1/\tanh \varepsilon > 1$; then $1/R < \tanh \varepsilon$, so $\operatorname{artanh}(1/R) < \varepsilon$, and Lemmas 1 and 2 give $d_K(p, q) \leq \varepsilon$. (Lean: `dK_eq_zero`.) $\square$

Theorem 6 (Lean: `C1305.main`, of type `C1305.Claim`). *Neither Reading A nor Reading B holds, and $d_K = 0$ on $D \times D$.*

Proof. Both readings require positive d_K between distinct points of D sharing a coordinate. The points $(i, 0), (i, 1)$ (first coordinate $i \notin \mathbb{Z}$) and $(0, i), (1, i)$ lie in D and are distinct, but Theorem 5 gives distance 0. This is exactly the elementary disc $z \mapsto (i, Rz)$. $\square$

4 Scope, conventions and audit

Conventions: d_K is the pseudodistance with the full Poincare distance and chains of arbitrary finite length; $d_K = \infty$ would be recorded if no chain existed (not relevant here). The reading “horizontal/vertical” is covered in both coordinate assignments. **Not formalised:** the infinitesimal Kobayashi–Royden metric; the same discs have derivative R in the w -direction at $(i, 0)$, so it also vanishes there, but this is argued informally only.

Lean audit: Lean 4.33.1 / Mathlib v4.33.1; the main theorem uses only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`, no `native_decide`. No literature is cited.